

in water. Tu wondered if this was the correct method for qinghao—and looked in reference materials, including Ge Hong's *A Handbook for Prescriptions for Emergencies*, written more than 1,600 years ago. He wrote, "A handful of qinghao, immersed in two liters of water, wring out the juice and drink it all." This made Tu think that heat from boiling the herb in a decoction might be destroying the active ingredients and that the aerial parts (the leaves and stalks) rather than the roots might have active properties, as these aerial parts could be wrung out. She therefore redesigned the experiment and extracted the active ingredients from the leaves and stems using a low-temperature method involving water, ethanol, and ethyl ether in 1971.

Her method worked, and Tu went on to isolate the active compound artemisinin, which successfully treated malaria in mice. Artemisinin kills the malaria parasite by damaging its membranes and interfering with the process by which it digests its host's hemoglobin. When analyzed chemically, artemisinin proved to be a sesquiterpene lactone, a different compound to any other known antimalarial drug.

Further success

Tu's team then had to carry out clinical studies on humans. In July 1972, she and two colleagues offered to evaluate

Tasked with finding a new antimalarial, Tu visited physicians across China practicing traditional herbal medicine. She collected more than 2,000 recipes within 3 months, including Ge Hong's (right) prescription for qinghao, which was key to Tu's successful discovery.

SATOSHI OMURA

Research by Japanese biochemistry professor Omura led to the creation of the drug ivermectin.



Omura (1935–) was a microbiologist at Kitasato University. Like Tu, he sourced medicines from natural substances, including the bacterium *Streptomyces avermitilis*. Further exploration of its potent antimicrobial properties eventually produced ivermectin, a key antiparasite medicine for many diseases, including river blindness and elephantiasis. Omura won the Nobel Prize in Physiology or Medicine in 2015 for his research.

artemisinin's toxicity by taking it themselves. They then ran a successful clinical trial in 1973, and the drug was licensed for use in 1986.

While continuing her research into artemisinin, Tu developed a second compound, dihydroartemisinin, that was 10 times more potent. This meant a smaller dose of the new compound would achieve the same effect. It was licensed by the Chinese Ministry of Health in 1992. Fast-acting artemisinin-based drugs are now widely given as part of combination therapy for malaria.

In 2011, Tu won the prestigious Lasker Award and, in 2015, she shared the Nobel Prize in Physiology or Medicine with Satoshi Omura and William C. Campbell, who had also used natural sources to create antiparasitic medicines. She was the first Chinese woman ever to win a Nobel Prize and the first Chinese person to win the Nobel Prize in Medicine. Today, Tu continues to work as chief scientist at the China Academy of Traditional Chinese Medicine.

**SCIENTISTS
WORLDWIDE
TESTED MORE THAN
240,000
COMPOUNDS
PRIOR TO THE
DISCOVERY OF
ARTEMISININ**



**RECEIVED THE
NOBEL
PRIZE
WHEN SHE
WAS
85**

**ARTEMISININ
COMBINATION
THERAPY TREATS
198 MILLION
PEOPLE
ANNUALLY**